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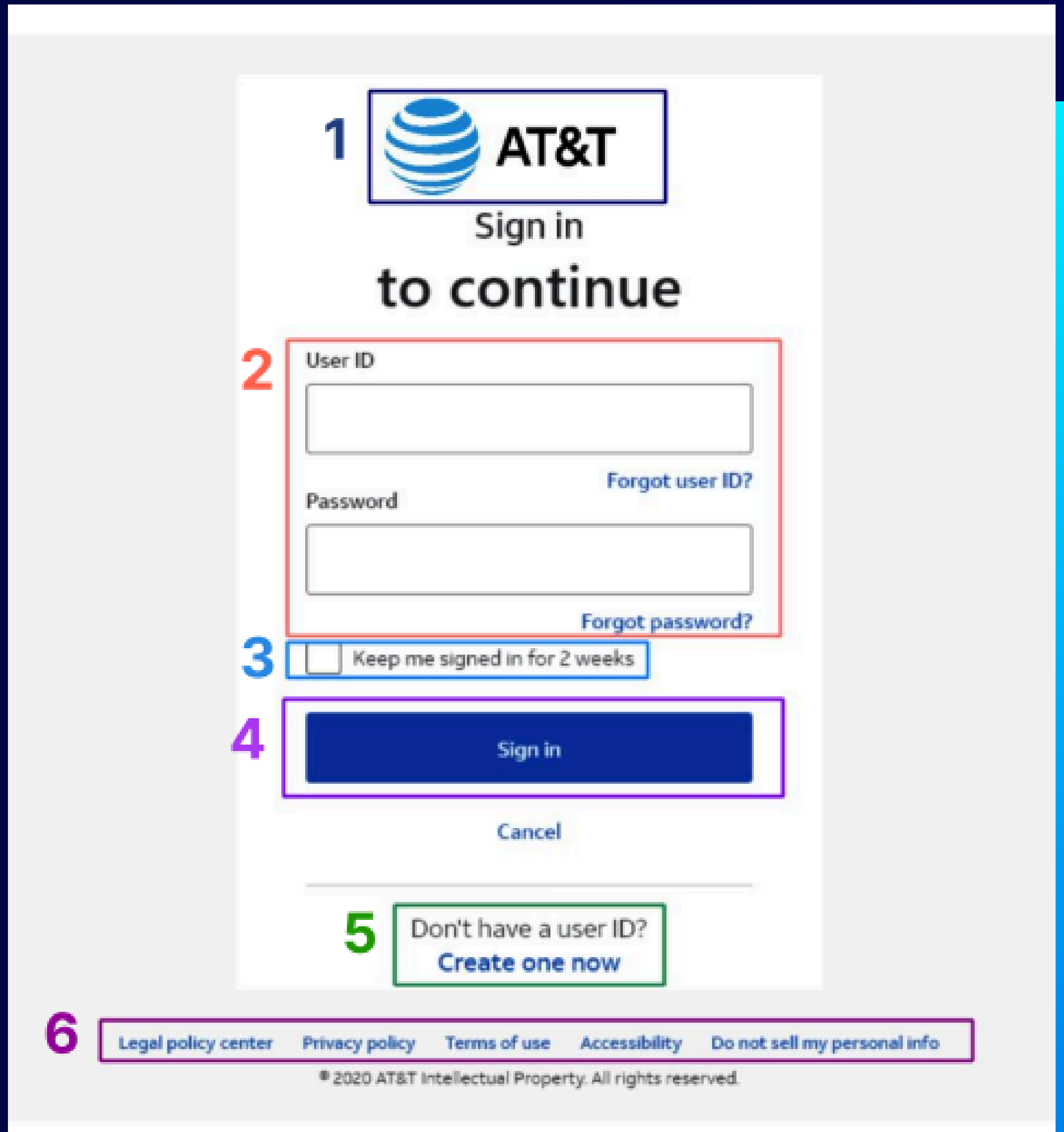
# PHISHLANG: A REAL-TIME, FULLY CLIENT-SIDE PHISHING DETECTION FRAMEWORK USING MOBILEBERT

[1]S. S. Roy and S. Nilizadeh, "PhishLang: A Lightweight, Client-Side Phishing Detection Framework using MobileBERT for Real-Time, Explainable Threat Mitigation," arXiv.org, 2024. Available: https://arxiv.org/abs/2408.05667




Original website

# PAPER'S METHOD

- **Core idea:** MobileBERT model on URL + source-code signals (multi-modal ensemble)
- **Parses website source code** – focuses on actionable tags (forms, inputs, scripts, links) instead of aesthetics
- **Multi-modal ensemble** – combines URL analysis + source code analysis
- **Efficient & client-side** – runs locally, protects privacy, needs less memory
- **Training dataset** – based on PhishPedia (30K phishing, 30K benign websites)

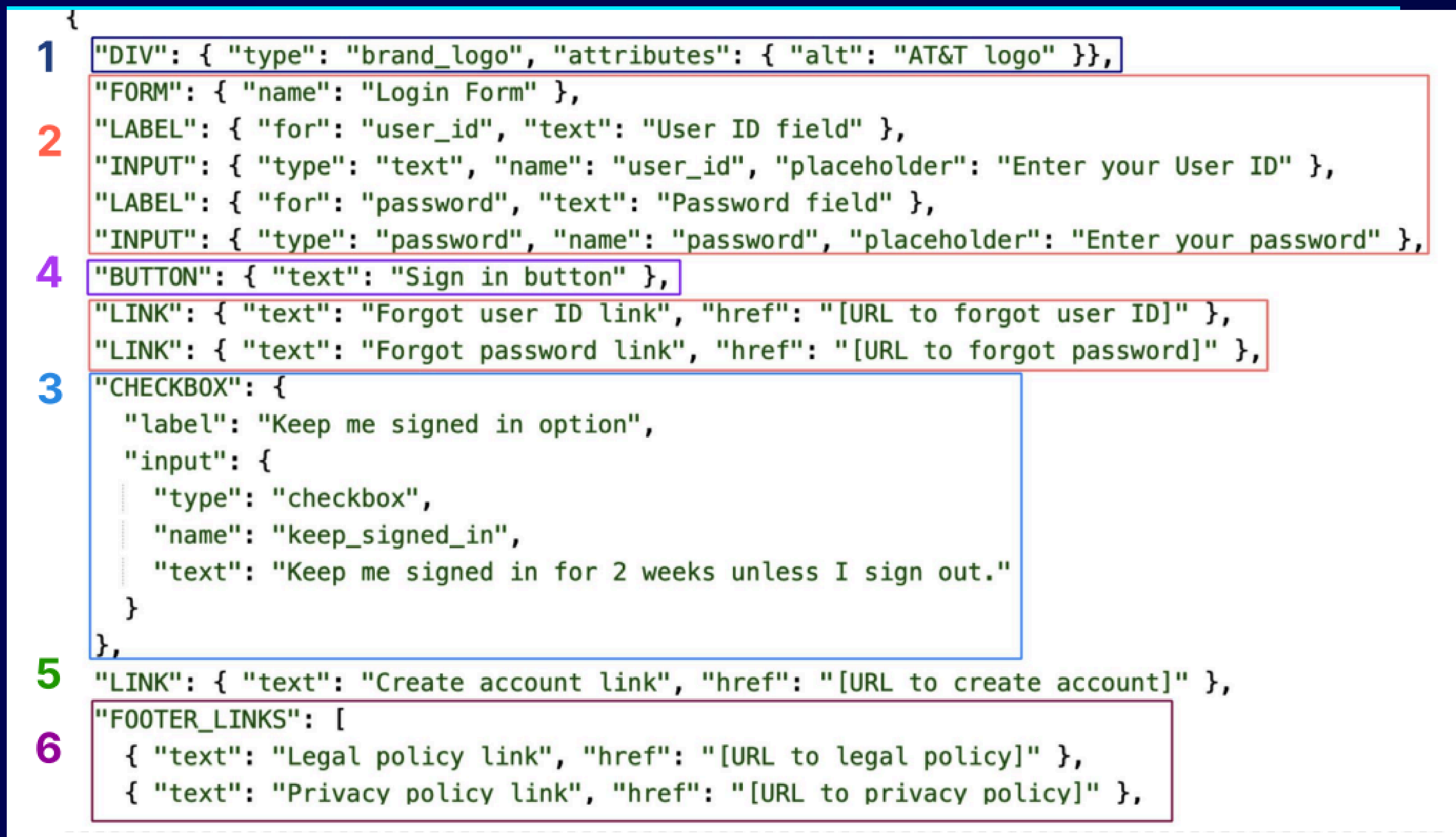
# MODEL CHOICE & KEY RESULTS

- **MobileBERT (lightweight, efficient, well-suited for client-side phishing detection)**

- **Chosen because:**
- Small footprint (~74 MB)
- Fast inference (<1s)
- Good accuracy on phishing/benign datasets

- **Key Results (Our Tests):**
- **Baseline (paper model reproduction):**
- Accuracy: 83.05%
- Precision: 71.93%
- Recall: 55.11%
- F1: 62.41%

- **Improved model (threshold τ=0.75):**
- Accuracy: 84.49% (+1.44 pp)
- Precision: 73.81% (+1.88 pp)
- Recall: 60.83% (+5.72 pp)
- F1: 66.70% (+4.29 pp)




Parsed representation

# HEAD-TO-HEAD METRICS

- The run with the **smaller accuracy (83.05%)** is the paper model reproduction, while our **improved version (threshold-tuned) reaches 84.49%**, with higher Precision, Recall, and F1.

| Model | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| Paper model (baseline) | 83.05% | 71.93% | 55.11% | 62.41% |
| Improved model (ours, optimal threshold = 0.75) | 84.49% | 73.81% | 60.83% | 66.70% |

```
==================================================
Starting Test Set Evaluation
==================================================
Test set: Average loss: 0.4314
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Metrics with Default Threshold (0.500):
  Accuracy: 80.80%
  Precision: 59.07%
  Recall: 80.73%
  F1 Score: 68.22%
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Metrics with Optimal Threshold (0.750):
  Accuracy: 84.49%
  Precision: 73.81%
  Recall: 60.83%
  F1 Score: 66.70%
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Evaluating on test set...
Test set: Average loss: 15.1257, Accuracy: 83.05%, Precision: 71.93%, Recall: 55.11%, F1: 62.41%
```

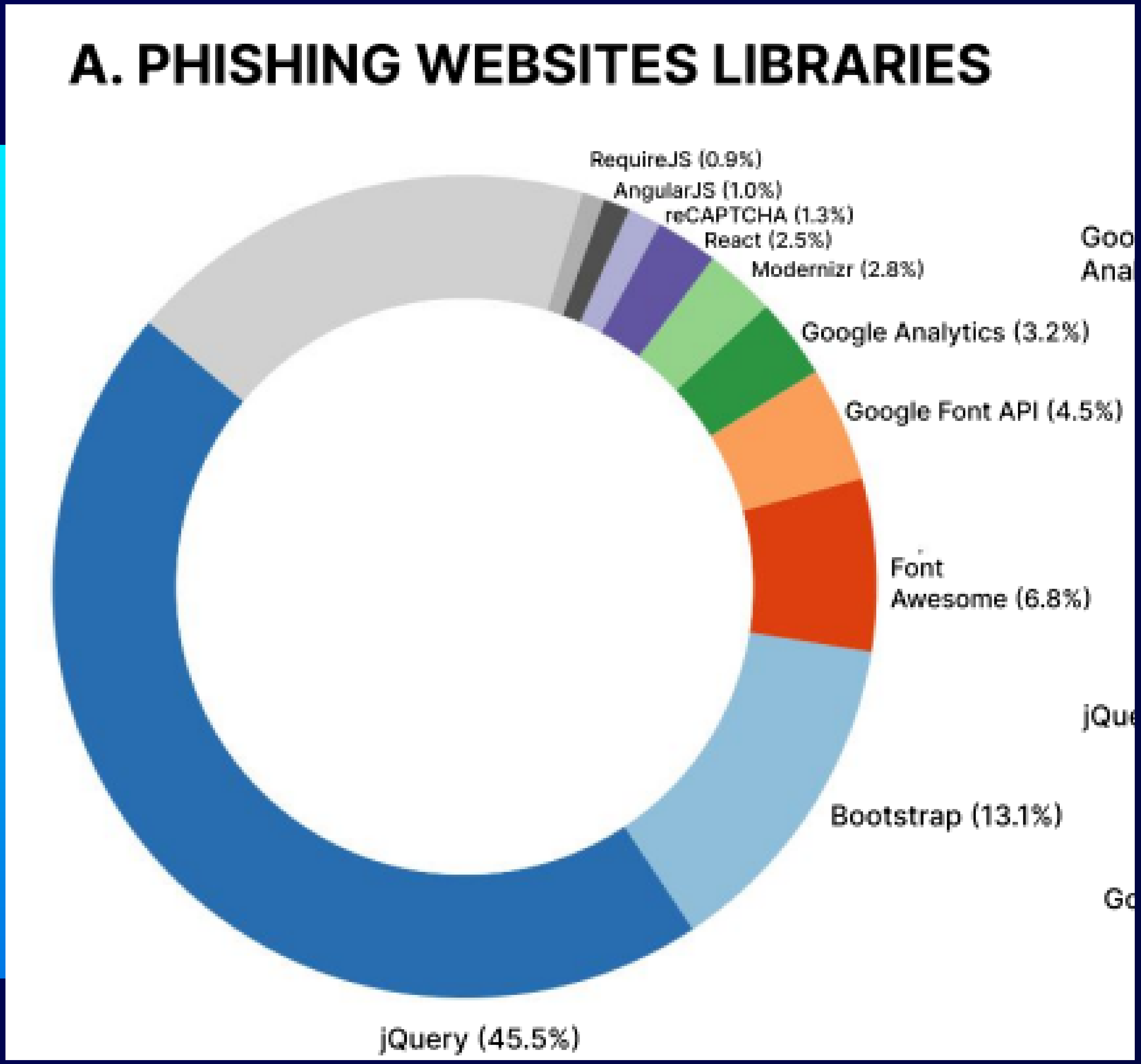
# GAP ANALYSIS

What they do better:

- **Multi-modal late-fusion:** Combine URL + parsed source-code, boosting robustness against obfuscation/zero-day tricks.
- **Client-side, real-time:** Chromium extension with local inference → privacy-preserving, sub-second typical latency.

What we do better:

- **Simpler & faster to iterate:** URL-only pipeline → easier training/deployment and rapid experimentation for the assignment.
- **Tuned decisioning = better scores:** Threshold τ=0.75 + consistent evaluation → +1.44 pp Accuracy and +4.29 pp F1 vs. our baseline reproduction.



# NEXT STEP

- Add FOIL dataset for training
- Faster convergence with richer samples
- Target: higher accuracy in next sprint
- Goal: Train the model faster and achieve higher accuracy by leveraging FOIL's curated phishing/benign samples.

THANK YOU VERY MUCH !!!